

Regression Criteria Reference

Throughout, y_i is the actual value for row i , $\hat{y}_i$ is the model's prediction for row i , n is the number of rows, x_{ij} is predictor j 's value for row i , α is the fitted intercept, and β_j is the fitted coefficient for predictor j , so that

$$\hat{y}_i = \alpha + \sum_j \beta_j x_{ij}.$$

1 OLS - Ordinary Least Squares

Minimizes:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The standard default regression. Fits coefficients with the closed-form solution $\hat{\beta} = (X^\top X)^{-1} X^\top y$. Squaring the residual means large errors are penalized much more than small ones, so a few big outliers can dominate the fit. Assumes errors are roughly Gaussian and of similar scale across all rows.

2 LAD/MAE - Least Absolute Deviations

Minimizes:

$$\sum_{i=1}^n |y_i - \hat{y}_i|$$

Minimizes absolute error instead of squared error. This is mathematically identical to fitting the conditional median of y given X , which is quantile regression at $\tau = 0.5$. Because outliers are not squared, a single extreme instance has much less influence on the fit than under OLS.

3 MAPE - Mean Absolute Percentage Error

Minimizes:

$$\frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Minimizes relative (percentage) error rather than absolute error. This matters when the target spans very different scales. MAPE treats a 5% gap the same everywhere, regardless of the instance's scale.

4 Huber - Huber loss (robust regression)

Minimizes:

$$\sum_{i=1}^n \rho_{\delta}(y_i - \hat{y}_i), \quad \rho_{\delta}(r) = \begin{cases} \frac{r^2}{2} & |r| \leq \delta \\ \delta \left(|r| - \frac{\delta}{2} \right) & |r| > \delta \end{cases}$$

A compromise between OLS and MAE: behaves like OLS (quadratic, smooth) for small residuals, but switches to behaving like MAE (linear) once a residual exceeds the threshold δ . This gives it OLS's statistical efficiency on well-behaved data while capping how much any single large outlier can drag the fit around, unlike pure OLS.

5 Quantile90 - Quantile regression at the 90th percentile

Minimizes:

$$\sum_{i=1}^n \rho_{\tau}(y_i - \hat{y}_i), \quad \rho_{\tau}(r) = \begin{cases} \tau r & r \geq 0 \\ (\tau - 1)r & r < 0 \end{cases}, \quad \tau = 0.9$$

Fits the conditional 90th percentile, i.e. a line that roughly 90% of actual costs fall at or below. It does this with an asymmetric loss: underpredicting ($y_i > \hat{y}_i$) is penalized $9\times$ more heavily than overpredicting ($\tau/(1 - \tau) = 0.9/0.1 = 9$), which pulls the fitted line up toward the upper end of the data's spread. Useful for a conservative upper bound estimate rather than a typical/average prediction.

Notes: Called “pinball loss” because its plot (loss vs. residual) looks like a V bent at the origin with unequal slopes on each side, like the profile of a pinball-machine lane.

6 Ridge - L2-penalized OLS

Minimizes:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_j \beta_j^2$$

OLS's squared-error objective, plus a penalty on the size of the coefficients themselves (their squared L_2 norm). This shrinks every coefficient toward zero (but never exactly to zero), which stabilizes the fit when predictors are correlated with each other. Plain OLS coefficients would otherwise be unstable/unreliable in that situation. α (penalty strength) is chosen automatically via 5-fold cross-validation.

7 Lasso - L1-penalized OLS

Minimizes:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_j |\beta_j|$$

Same idea as Ridge, but penalizes the absolute value of coefficients instead of their square. It can drive unhelpful predictors' coefficients to exactly zero, effectively removing them from the model.

8 LogLinear - OLS on $\log(y)$

Minimizes:

$$\sum_{i=1}^n (\log y_i - X_i \beta)^2, \quad \text{then predicts: } \hat{y}_i = \exp(X_i \beta)$$

Fits a straight line to the logarithm of the cost and then exponentiates the predictions back to the original scale. This makes it a multiplicative model (each predictor scales the cost by a percentage, rather than adding/subtracting a fixed amount), which can be a better fit than a plain linear scale for strictly-positive data that spans a wide range.

9 Log-Log - OLS on $\log(y)$ vs. $\log(X)$

Minimizes:

$$\sum_{i=1}^n (\log y_i - \log(X_i) \beta)^2$$

Unlike LogLinear above, this logs the predictors as well as the target, so each β_j is an elasticity (a 1% change in x_j moves y by roughly $\beta_j\%$) rather than a linear slope. Skipped if any y_i or predictor value is not strictly positive. Best suited to raw, scale-driven predictors (e.g. `sqrt_An`, `LB`, `Average_inv`), where it clearly outperforms every other criterion.